Show your work. All problems are worth 4 point unless stated otherwise.

1. Write the expression, $x + x^2 + x^3 + x^4 + \cdots + x^{20}$ in sigma notation.

2. Write out the following sum $\displaystyle\sum_{n=0}^{3} n^2$.

Evaluate the geometric series or state that it diverges.

3. $\displaystyle\sum_{k=0}^{\infty} \frac{1}{4^k}$

4. $\displaystyle\sum_{m=0}^{\infty} \frac{(-3)^{2n}}{5^n}$

Using the Squeeze Theorem, find the limit of the following sequence or determine that the limit does not exist.

5. $\displaystyle\lim_{n\to\infty} \left\{ \frac{(-1)^n}{n} \right\}$

Use the Divergence Test to determine whether the following series diverge, or state that the test is inconclusive.

6. $\displaystyle\sum_{k=0}^{\infty} \frac{1}{k+1}$

7. Let $f(x) = 3e^x$. (Each part is worth 4 points.)

a. Find the Taylor polynomial of order $n = 2$ for $f(x)$ centered at 0.

b. Use the Taylor polynomial from above to approximate $3e^{0.1}$, and compute the absolute error in the approximation assuming the exact value is given by a calculator.

Give a set of parametric equations that describes the curve.

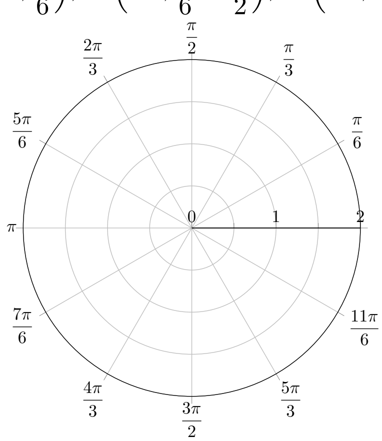
8. Give two different sets of parametric equations that generate a circle centered at $(0, 0)$ with radius 5.

a. [2 pts] Set one:

b. [2 pts] Set two:

Locate the following polar coordinates on a graph.

9. A:$\left(1.5, \dfrac{\pi}{6}\right)$, B:$\left(1.5, \dfrac{\pi}{6} + \dfrac{\pi}{2}\right)$, C:$\left(1.5, -\dfrac{\pi}{6}\right)$, D:$\left(-1.5, -\dfrac{\pi}{6}\right)$



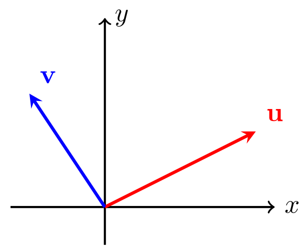
Express the following ***polar*** *coordinates in* ***Cartesian*** *coordinates.*

10. $\left(2, \frac{\pi}{3}\right)$

Use the graph to draw the following vectors (provide labels). Make certain to indicate direction.

11. **a.** [2 pts] $\frac{1}{2}\mathbf{v}$

b. [2 pts] $\mathbf{u} + \mathbf{v}$



Use the vectors $\mathbf{u} = \langle -1, 1, 1\rangle$ *and* $\mathbf{v} = \langle 2, -4, 0\rangle$ *to complete the following problems.*

12. $\mathbf{u} \cdot \mathbf{v}$

13. Find the angle between $\mathbf{u}$ and $\mathbf{v}$. ($|\mathbf{u}| = \sqrt{3}$ and $|\mathbf{v}| = 2\sqrt{5}$.)

14. Find a vector perpendicular to $\mathbf{u}$ and $\mathbf{v}$.

15. A woman paddles a canoe at 5 mi/hr at 30 degrees ($\frac{1}{6}\pi$ radians) north of east. Find a vector describing the vertical and horizontal components of the canoe's velocity.

Find the indicated partial derivative of the following functions.

16. Find f_x and f_y for $f(x, y) = 5x \cdot \sin 2xy$.

17. Find the four second partial derivatives of $f(x, y) = 2x^5y^2 + 2x^2$.

Use the Chain Rule to find the following derivative. When feasible express your answer in terms of the independent variable.

18. dW/dt where $W = x^2 + y^2$, $x = \cos 2t$, and $y = \sin 2t$.

Compute the gradient of the following function and evaluate it at the given point P.

19. $f(x, y) = e^{2xy}$; $P(2, 3)$

Let $f(x, y) = \ln(xy)$ and $P(2, 3)$ be a point on f.

20. Find a vector that points in a direction of steepest ascent in f at P.

21. Find a vector that points in a direction of no change in f at P.

Use the Second Derivative Test to determine (if possible) whether each critical point of the given function is a local maximum, local minimum, or saddle point.

22. [8 pts] $g(x, y) = x^4 + y^4 - 16xy$ with critical points $(0, 0)$, $(2, 2)$, and $(-2, -2)$.

BONUS

Use Lagrange multipliers to find the maximum and minimum values of f (when they exist) subject to the constraint.

23. $f(x, y) = e^{2xy}$ subject to $x^3 + y^3 = 16$